

Measuring a Laser Beam Profile with Optical Lenses

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Lasers are crucial components in our everyday life that have countless applications in our modern society. Understanding how lasers operate is important in developing new applications and understanding their limitations. In this lab, we aim to explore the transverse beam profile of a laser by measuring its intensity as a function of position in one dimension. By expanding and collimating the beam using optical lenses, we will graph the intensity profile of the expanded beam. This experiment will deepen our understanding of Gaussian beam profiles and the principles of optical lenses.

The setup to complete this task was composed of a laser whose beam is expanded using a Newtonian telescope and reflected onto a detector, as shown in Fig. a. We used lenses with focal lengths of 100 mm and 400 mm to achieve a 4x magnification. To measure the beam profile of the laser, a razorblade mounted on a sliding stage was utilized to measure the change in intensity of the laser as measured by the detector across different positions of the stage, from full transmission to completely blocked by the razorblade. By measuring the position of the razorblade (and thus, the portion of the laser that was blocked) and the detector reading, a graph can be composed measuring intensity as a function of position in 1-D. This relationship was then fitted onto a complementary error function $G(x) = I \times \text{erfc}(\sqrt{2}(x - x_0)/w) + I_b$, where I is the beam intensity, I is the relative integrated intensity in arbitrary units, w is the beam width, x_0 is the offset position, and I_b is the background intensity (Fig. b). Its numerical derivative was also fitted onto a Gaussian function $g(x) = I_0 \exp(-2(x - x_0)^2/w^2)$ to find the beam radius before and after the beam expander (Fig. c). Additionally, a secondary Gaussian curve is considered composed of interpolated data consisting of 100 equidistant points substituting original data.

Table 1 shows the results of the different fits to calculate the beam radius as well as the effective magnification. w_i is the beam radius before expansion, and w_f represents the same quantity after it is expanded. m is the factor of magnification.

	w_i (mm)	w_f (mm)	m
Error function fit	0.261 ± 0.0023	1.028 ± 0.013	3.94 ± 0.061
Gaussian function fit	0.261 ± 0.0012	1.044 ± 0.022	4.01 ± 0.086
Gaussian function fit with interpolated data	0.264 ± 0.0005	1.030 ± 0.0009	3.92 ± 0.008

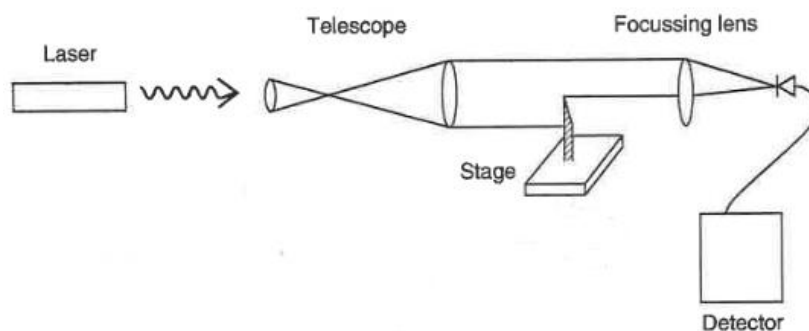


Figure a: Beam expander setup using a Newtonian telescope to measure the transverse beam profile of a laser. The focal length of the first mirror (left) is 100mm and the second (right) is 400mm. The setup allows for 4x magnification.

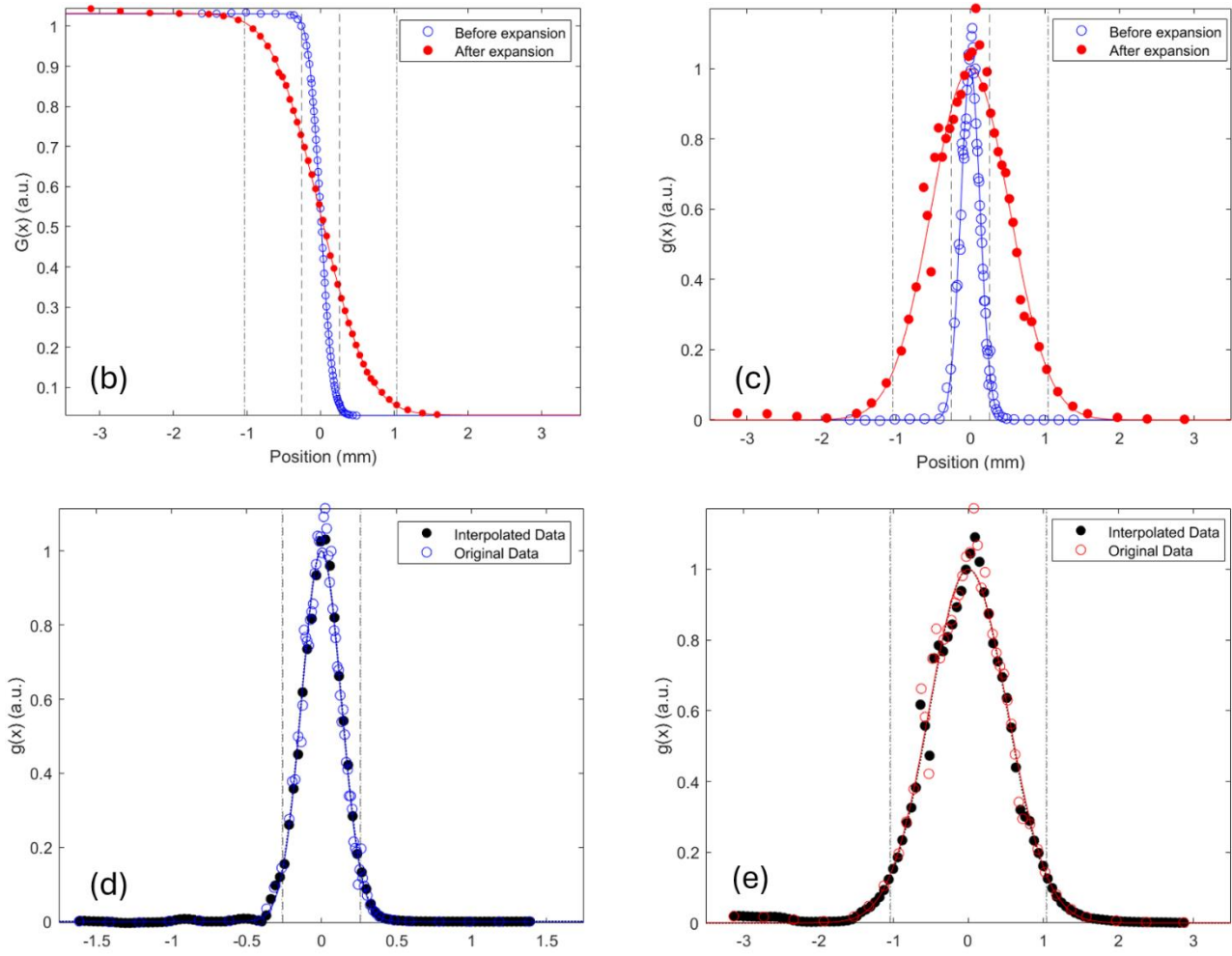


Figure b: Error function fitting before and after the beam expansion. Dashed vertical lines visually indicate the beam radius, w_0 . Before the expansion, $w_i = 0.2610$ mm. After the expansion, $w_f = 1.0284$ mm. The effective beam expansion is 3.94x. Figure c: Gaussian function fitting before and after the beam expansion. Dashed vertical lines visually indicate the beam radius, w . Before the expansion, $w_i = 0.2605$ mm. After the expansion, $w_f = 1.0441$ mm. The effective beam expansion is 4.01x. Figure d: Gaussian fit before expansion using interpolated data. Using this data, $w_i = 0.2627$ mm. Figure e: Gaussian fit after expansion using interpolated data. Based on this data, $w_f = 1.0296$ mm. For both interpolation fits, 100 equidistant points were plotted.